

Research Article

Mathematical Modeling and Analysis on the Population Birth Rate Trends in Japan

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Japan faces a significant demographic challenge with far-reaching social, economic, and political ramifications due to its declining birth rate and rapidly rising aging population. Addressing the intricate aspects of these serious emergent phenomena, this study has proposed a mathematical model using a system of ordinary differential equations that has been observed analytically and numerically. As part of the analytical solution, we have verified the model's existence and uniqueness, investigated its stability at the equilibrium points, and explored its characteristics concerning different state variables. Besides, a sensitivity analysis has been conducted to identify the sensitive parameters based on the estimated parameters. Additionally, the Runge–Kutta fourth-order approach has been used to find the numerical solutions. This study's findings have been investigated using a newly proposed mathematical model to highlight the demand for a broad awareness campaign in Japan to address the possible effects of a declining birth rate on social structure and the country's economy, as well as concerns about rising prices influencing family planning decisions. To sustain population growth in Japan, the government should provide proper initiatives to boost the birth rate.

Keywords: demographic profile; incentives; stability; well-posedness; workaholism

1. Introduction

A dramatic decrease in birth rates and an aging population have altered Japan's demographic profile in recent decades. Numerous areas, including healthcare, labor force dynamics, and pension systems, are affected by this phenomenon, which presents substantial difficulties to the nation's socioeconomic fabric. Policymakers, academics, and stakeholders who are seeking effective methods to address these demographic transitions must have a deep understanding of the complexities of population birthrate patterns.

Japan is well renowned globally and boasts an impressive amount of records. Besides, Japan has recently achieved a new milestone in terms of declining birth rate and obtained the title of the 'Nation of the Elderly' [2]. Following 2008, however, the population of Japan has been consistently declining (refer to Figure 1). This trend is also reflected by a negative continuing growth rate, as illustrated in Figure 2.

In comparison to other postindustrial countries, Japan's population is aging at a faster rate. As a whole, the average number of children born to each woman, or the birth rate, started fall in the 1970s. The current birth rate is 1.3 and is much lower than the "replacement level" of slightly more than two children per woman, which is often considered necessary to sustain a steady population [4]. Twenty percent of Japanese men and fifteen percent of Japanese women had no interest in getting married in a 2022 survey by Japan's National Institute of Population and Social Security Research as described in Raymo, Uchikoshi, and Yoda [5]. These numbers are the best levels seen since 1982. Thirty percent of Japanese men and twenty percent of Japanese women in their fifties have never gone into a marriage. Also, day by day worries for Japan's economy are rising as a result of a lack of people of working age and youngsters who will soon join labor [6]. Furthermore, in Japan, a significant number of supermarkets are shutting down, which presents

challenges for the senior population in obtaining necessary foods and everyday products as a result of the population reduction [7] because the demographics of Japan indicate to a dwindling population as well as an aging population. As a result of a predicted decline in the number of young adults, the percentage of the population that is of working age will fall and the average age of the population would rise [8]. While a smaller population means decreased employment and less production, increasing capital intensity, productivity gains, and labor participation rates could reduce the blow on incomes per person [9]. Besides, the Japanese government announced the numbers in observance of Respect for the Aged Day, a national holiday [10]. A diminishing labor force due to a decreasing birth rate can also have an effect on health care and pension financing due to rising demand from an older population [11]. Japan is not the only country facing this problem; nearby Taiwan, China, Singapore, and South Korea are all fighting a losing battle to persuade young people to have more children despite rising living expenses and a fractured society [12]. Meanwhile, the Japanese Ministry of Justice has documented a steady rise in the number of foreign residents since World War II as a result of economic expansion, scarcity of labor for rising older population, demographic issues, policy changes, globalization, educational prospects, and initiatives to draw in international expertise [13]. For instance, more foreign nationals with resident permits entered Japan than departed, and it does not even include those on temporary visas. In 2020, 183,600 people were a net result of 733,600 people entering and 550,000 people leaving [14]. Besides, by 2022, Japan registered over 3.1 million foreign nationals residing in the country, representing less than 3 percent of the total population. Consequently, immigration can also play a significant role in the variation of Japan's population by increasing its contribution day by day (see Figure 3).

Furthermore, Rahman [16] reported that the government of the nation of Japan is introducing a range of policies and incentives to alleviate the burden on Tokyo. As part of its initiatives, the organization provided families who were relocating from the Tokyo metropolitan region with financial assistance of up to 1 million yen per kid after last year. The implementation of this new regulation will commence in April 2023. A family consisting of two parents and two children below the age of 18, who establish a company in a different locality, will be granted E 500,000 as monetary aid by the government. The Japanese government intends to provide free medical treatment to school-children until they reach the age of 8. Besides, to better understand how natural events affect groups, areas, and countries, mathematical models have become useful tools for analysis [17]. Again, the birth rate in Singapore has just reached a historic low, dropping from 1.12 in 2021 to 1.04 in 2022. In 2022, South Korea and Japan both recorded low birth rates of 0.78 and 1.2, respectively. Although these nations are wealthy and advanced with abundant resources, they have not achieved success in increasing birth rates via financial incentives such as cash bonuses and paid parental leave [18]. However, incentives can be failed for different reasons to address the problem. One of them, the

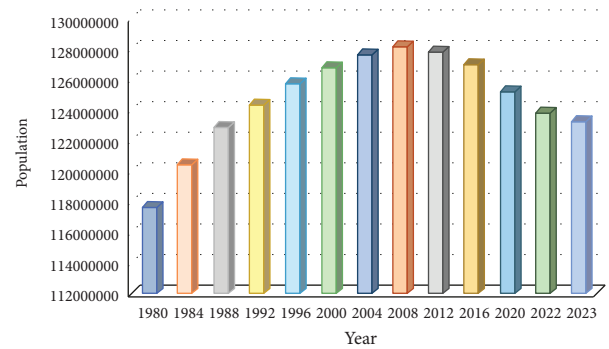


FIGURE 1: A representation of the population of Japan in graphical form [1].

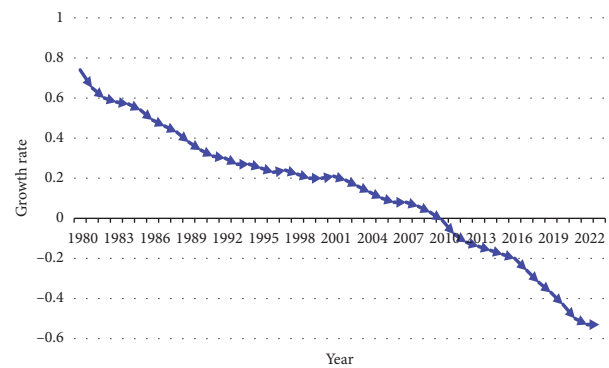


FIGURE 2: Japan's population growth rate is falling day by day [3].

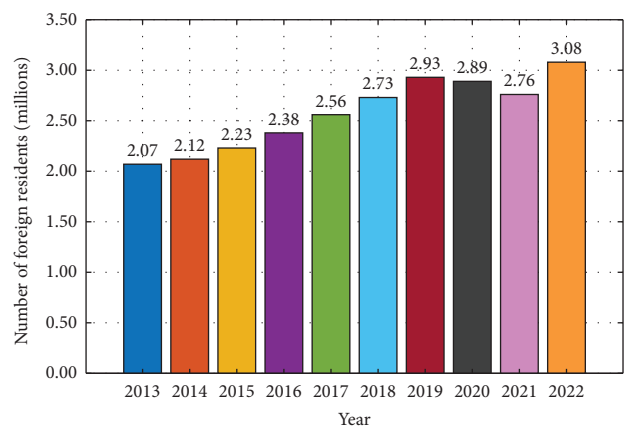


FIGURE 3: Number of registered foreign residents through immigration in Japan from 2013 to 2022 [15].

unwillingness of some Japanese people to establish a family may be explained by several factors. Firstly, because of the prevalent work culture, which is defined by high job expectations and long working hours, family life is frequently neglected in favor of profession [19]. Further discouraging people from contemplating motherhood are economic concerns with price inflection, which include high housing and living costs. Extra demands and disincentives to plan a family could come from social and cultural sources including long-established gender norms and cultural

expectations. Couples are putting off making choices about family planning and getting married later in life, which is adding to the demographic crisis [20]. Also, a number of factors have influenced family relations as a result of technology's ubiquitous presence in Japanese society. Digital communication tools and social media platforms are everywhere, changing the way people communicate with one another and maybe leading to fewer in-person community gatherings. Isolation may set in as a result of people spending more time in front of screens and less time interacting with one another, which threatens the social networks that are vital to family life [21].

On this track, Usman and Tomimoto [22] studied Japanese youth's future field selecting study year. After investigation, some regulations have been suggested to provide flexibility and encouragement to private sector women with children, increase baby daycare and elderly houses, and launch informative courses on the scope and future job opportunities in different fields in the second and third grades of high school. Higazy et al. [23] elucidated the optimal control strategy to mitigate the fatal Lassa Hemorrhagic Fever Disease (LHF) disease that is caused in pregnant women and poses a significant threat to the escalating birth rate. On the other side, disseminating information via traditional forms of media such as newspapers, magazines, television, and radio may effectively raise public awareness of the ongoing problem [24].

After analyzing the available literature, we have considered the birth rate as a variable for model formulation that can potentially impact the behavior of Japan's population in our study. Thus, this study presents a mathematical model based on a system of ordinary differential equations to investigate the fluctuation of birthrate in Japan and provide strategies to mitigate the decline in population growth, ensuring sustained development with a focus on the young population.

The subsequent section of the article is organized in the following manner: The mathematical model for this problem has been formulated in Section 2. Section 3 establishes the well-posedness of the model, while Sections 4 and 5 include a study of the equilibrium, including stability and other qualitative aspects. Section 6 presents the estimates of the parameters. After, Section 7 demonstrates the sensitivity of the parameter based on the values obtained in Section 6. Following this, the numerical simulation, together with its corresponding commentary, has been presented in Section 8. Subsequently, Section 9 presents the findings together with corresponding remarks. The last piece, Section 10, presents the findings and outcomes of this investigation.

2. Derivation of Mathematical Model

In this section of the study, we are going to develop an interesting mathematical model, drawing on existing reports, that reliably reflects the population's dynamics concerning the variation of the birth rate.

For the development of our model, we have incorporated two state variables: the population in Japan, denoted as $P(t)$, and the birth rate, denoted as $B(t)$. Additionally, our model

formulation assumes that the population in Japan is solely influenced by the birth rate.

Population growth is often attributed to an increase in the birth rate. At the same time, the population of a country increases as a result of immigration from another nation, as individuals relocate to the country in search of diverse prospects for their daily lives. Meanwhile, the expansion of this phenomenon is hindered by the loss of resources caused by the crowding effects resulting from intraspecies competitions. At the same time, the population of a country also declines as a result of increasing natural death. With this concept in consideration, we can express our first differential equation as

$$\frac{dP}{dt} = \alpha BP - \gamma P^2 - \mu P, \quad (1)$$

with $\gamma = r/K$.

Where α is the increasing rate of a population based on by immigration and birth rate, r is the competition coefficient, and carrying capacity is K . Here, the natural death rate is represented by μ .

Again, the birth rate plays an indispensable part in determining a country's population improvement. It is demonstrated that public awareness campaigns regarding the negative impacts of population decline on national development may increase the birth rate. The Japanese, meanwhile, are too preoccupied with their careers and technical advancements to give any thought to starting a family. Additionally, keeping up with the cost of raising a family requires a substantial sum of money, which is further complicated by price inflation. Regarding this matter, the government may enhance the birth rate by providing incentives to individuals to have more children. Based on these issues, our second governing equation will be formulated as

$$\frac{dB}{dt} = \tau + \sigma B - \xi B - \omega B - \delta B, \quad (2)$$

where the parameters τ and σ indicate the increase in birth rate resulting from the promotion of awareness and the provision of incentives, respectively. The decline in birth rate caused by work business, technical advancements, and price inflation is denoted by ξ , ω , and δ accordingly.

Finally, our model equation will be as follows:

$$\frac{dP}{dt} = \alpha BP - \gamma P^2 - \mu P, \quad (3)$$

$$\frac{dB}{dt} = \tau + \sigma B - \xi B - \omega B - \delta B, \quad (4)$$

with initial conditions $P(0) = P_0 \geq 0$ and $B(0) = B_0 \geq 0$ and Figure 4 displays the diagram of this model.

3. Well-Posedness of the Model

To establish the mathematical and biological significance of the proposed model (3) and (4), it is necessary to provide proof of the existence, non-negativity, and boundedness of its solutions.

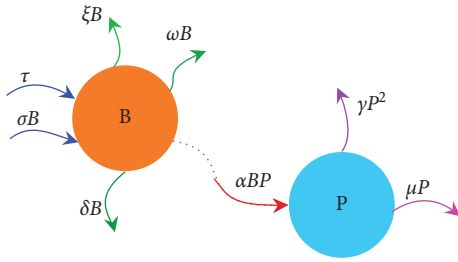


FIGURE 4: Schematic diagram of the proposed model.

3.1. Non-Negativity of the Model Solution

Lemma 1. If $P(0) \geq 0$ and $B(0) \geq 0$, then $P(t)$ and $B(t)$ will be always positive for all $t \in [0, T]$ in $\mathbb{R}_+^2$ where $T > 0$.

Proof 1. Taking all parameters of the system and all initial values to be positive, we have to prove that $P(t)$ and $B(t)$ will be positive for all $t \in [0, T]$ in $\mathbb{R}_+^2$.

From equation (4), we can write as follows:

$$\begin{aligned} \frac{dB}{dt} &= \tau + \sigma B - \xi B - \omega B - \delta B \\ \implies \frac{dB}{dt} &\geq B(\sigma - \xi - \omega - \delta) \\ \implies \frac{dB}{B} &\geq (\sigma - \xi - \omega - \delta)dt. \end{aligned} \quad (5)$$

Integrating both sides we get

$$\implies \ln B \geq (\sigma - \xi - \omega - \delta)t + \ln c. \quad (6)$$

Then this result will show as

$$\begin{aligned} \implies B &\geq B_0 e^{(\sigma - \xi - \omega - \delta)t} \\ \therefore B(t) &\geq B_0 e^{(\sigma - \xi - \omega - \delta)t} > 0. \end{aligned} \quad (7)$$

Therefore, we have found that $B(t) > 0$. Again, equation (3) gives,

$$\begin{aligned} \frac{dP}{dt} &= \alpha BP - \gamma P^2 - \mu P, \\ \implies \frac{dP}{dt} &= (\alpha_1 B - \gamma P - \mu)P, \\ \implies \frac{dP}{P} &= (\alpha_1 B - \gamma P - \mu)dt. \end{aligned} \quad (8)$$

After integrating both sides, we get

$$\begin{aligned} \ln P &= \alpha_1 \int B dt - \mu t - \gamma \int P dt, \\ \therefore P(t) &= e^{-\mu t + \int (\alpha_1 B - \gamma P) dt} > 0. \end{aligned} \quad (9)$$

These all complete our proof. $\square$

3.2. Model Solution's Boundedness

Lemma 2. All the solution trajectories of the model (3) and (4) are constrained inside certain boundaries and attract all solutions originating inside the positive quadrant's interior, that is the biological feasible region for system (3) and (4) is defined by the set Ω

$$\Omega = \left\{ (P(t), B(t)) \in \mathbb{R}_+^2 : P(t) + B(t) \leq \frac{\nu}{\Lambda}, t \geq 0 \right\}. \quad (10)$$

Proof 2. Let us consider that, $N(t)$ represents the summation of the solutions $P(t)$ and $B(t)$ in our proposed model. That means,

$$N(t) = P(t) + B(t),$$

$$\implies \frac{dN}{dt} = \frac{dP}{dt} + \frac{dB}{dt}, \quad (11)$$

$$\implies \frac{dN}{dt} + \Lambda N = \frac{dP}{dt} + \frac{dB}{dt} + \Lambda(P + B).$$

Putting the values we get

$$\begin{aligned} \therefore \frac{dN}{dt} + \Lambda N &= \alpha BP - \gamma P^2 - \mu P + \tau + \sigma B - \xi B - \omega B - \delta B \\ &\quad + \Lambda P + \Lambda B. \end{aligned} \quad (12)$$

Let us now have a look at a function of $h(x, y)$ that is put out in the following manner:

$$\begin{aligned} h(x, y) &= \alpha xy - \gamma x^2 - \mu x + \tau + \sigma y - \xi y - \omega y - \delta y \\ &\quad + \Lambda x + \Lambda y, \end{aligned} \quad (13)$$

where (x, y) is a critical point.

For our purpose we assume that, $W = h_{xx}h_{yy} - h_{xy}^2$.

Substituting $h_{xx} = -2\gamma$, $h_{yy} = 0$ and $h_{xy} = \alpha$, we obtain

$$W = -(-2\gamma)(\alpha) = 2\gamma\alpha > 0. \quad (14)$$

Due to the fact that $W > 0$ and $h_{xx} < 0$ with $h_{yy} = 0$, $h(x, y)$ holds a local maximum. Hence, According to the information mentioned in Kundu and Mallick [25]; taking the maximum value of ν to be taken into consideration equation (12) states that,

$$\frac{dN}{dt} + \Lambda N \leq \nu. \quad (15)$$

Taking the limit supremum, we obtain

$$\lim_{t \rightarrow \infty} \text{Sup}\{N(t)\} \leq \frac{\nu}{\Lambda}. \quad (16)$$

This leads us to the conclusion that the system's $P(t)$ and $B(t)$ states are all bounded. $\square$

3.3. Existence and Uniqueness of Solution

Lemma 3. Let M be the domain that satisfies the Lipschitz criteria. The system's solutions exist and are unique for every time $T \geq 0$ in domain D for all non-negative initial conditions.

Proof 3. The theorem described in Sowole et al. [26] has shown that it is necessary to adhere to the suggested Lipschitz criteria in order to ensure the existence and uniqueness of a solution inside a specific region, denoted as M . Let

$$\begin{aligned} f(P, B) &= \frac{dP}{dt} = \alpha BP - \gamma P^2 - \mu P, \\ g(P, B) &= \frac{dB}{dt} = \tau + \sigma B - \xi B - \omega B - \delta B. \end{aligned} \quad (17)$$

Using above system's equation, the partial derivative of f, g , with respect to compartments P, B are obtained as

$$\frac{\partial f}{\partial P} = \alpha B - 2\gamma P - \mu, \left| \frac{\partial f}{\partial P} \right| = |\alpha B - 2\gamma P - \mu| \leq \frac{\nu}{\Lambda} < \infty, \quad (18)$$

$$\frac{\partial f}{\partial B} = 0, \left| \frac{\partial f}{\partial B} \right| = 0 \leq \frac{\nu}{\Lambda} < \infty.$$

Again,

$$\frac{\partial g}{\partial P} = 0, \left| \frac{\partial g}{\partial P} \right| = 0 \leq \frac{\nu}{\Lambda} < \infty, \quad (19)$$

$$\frac{\partial g}{\partial B} = \sigma - \xi - \omega - \delta, \left| \frac{\partial g}{\partial B} \right| = |\sigma - \xi - \omega - \delta| \leq \frac{\nu}{\Lambda} < \infty.$$

Therefore, we have shown that all the partial derivatives exhibit continuity and boundedness. Consequently, the requirements set by Lipschitz condition are fulfilled. Therefore, according to the theory presented in Sowole et al. [26]; there is only one solution for system (3) and (4) inside the region M .

However, considering subsections 3.1–3.3, we may affirm that our presented model is well-posed.

4. Equilibria Analysis With Its Stability

4.1. Equilibrium Points. We can determine the point of equilibrium by solving the equations given below.

$$f(P, B) = \alpha BP - \gamma P^2 - \mu P = 0, \quad (20)$$

$$g(P, B) = \tau + \sigma B - \xi B - \omega B - \delta B = 0. \quad (21)$$

Solving equations (20) and (21), we get two types of non-negative equilibrium points E_1 and E_2 .

- The axial equilibrium point $E_1 = (0, \tau/\kappa)$ exists on the boundary of the first quadrant if $\delta + \omega + \xi > \sigma$ where $\kappa = \delta - \sigma + \omega + \xi$ and

- The positive interior equilibrium point $E_2 = ((\alpha\tau - \mu\kappa/\gamma\kappa), \tau/\kappa)$ exists if $\alpha\tau + \mu\sigma > \mu(\xi + \omega)$ and $\delta + \omega + \xi > \sigma$ where $\kappa = \delta - \sigma + \omega + \xi$

4.2. Future Status of Japan Population. Using the next generation matrix method Diekmann et al. [27]; we would find the upcoming status of population R_0 in Japan. To examine the dynamical behavior of human population in Japan, the state variable P has been taken for the discussion. We take

$$\frac{dP}{dt} = \alpha BP - \gamma P^2 - \mu P. \quad (22)$$

Differentiating it with respect to P , we get

$$\implies \frac{d}{dP} \left(\frac{dP}{dt} \right)_{P=P^*, B=B^*} = \alpha B^* - (2\gamma P^* + \mu). \quad (23)$$

Therefore, two matrices M and D which represent increase and decrease of Japan population are $M = \alpha B^*$ and $D = 2\gamma P^* + \mu$.

Then the future situation of Japan's population would be R_0 will be

$$R_0 = \frac{M}{D} = \frac{\alpha B^*}{2\gamma P^* + \mu}. \quad (24)$$

Consequently, the population in Japan will be boosted if $\alpha B^* > 2\gamma P^* + \mu$. In contrast, the population lessens if $\alpha B^* < 2\gamma P^* + \mu$.

However, at the interior equilibrium point, the future status of the population in Japan can be written as

$$R_0 = \frac{\alpha(\tau/\kappa)}{2\gamma(\alpha\tau - \mu\kappa/\gamma\kappa) + \mu}. \quad (25)$$

4.3. Local Stability

Theorem 1 (Local Stability Theorem). The dynamical system is unstable at equilibrium point E_1 .

Proof 4. To proof Theorem 1, at first we consider

$$f(P, B) = \alpha BP - \gamma P^2 - \mu P, \quad (26)$$

$$g(P, B) = \tau + (\sigma - \xi - \omega - \delta)B. \quad (27)$$

For equations (26) and (27), the Jacobian matrix is

$$J = \frac{\partial(f, g)}{\partial(P, B)} = \begin{pmatrix} \frac{\partial f}{\partial P} & \frac{\partial f}{\partial B} \\ \frac{\partial g}{\partial P} & \frac{\partial g}{\partial B} \end{pmatrix}, \quad (28)$$

$$\implies J = \begin{pmatrix} \alpha B - 2\gamma P - \mu & \alpha P \\ 0 & \sigma - \xi - \omega - \delta \end{pmatrix}. \quad (29)$$

At the equilibrium point $E_1 = (0, \tau/\kappa)$, equation (29) becomes

$$\Rightarrow J|_{E_1} = \begin{pmatrix} \alpha \frac{\tau}{\kappa} - \mu & 0 \\ 0 & \sigma - \xi - \omega - \delta \end{pmatrix}, \quad (30)$$

$$\therefore J|_{E_1} = \begin{pmatrix} \alpha \frac{\tau}{\kappa} - \mu & 0 \\ 0 & -\kappa \end{pmatrix},$$

where $\kappa = \xi + \omega + \delta - \sigma$

So, the characteristic equation is

$$|J|_{E_1} - \lambda I| = 0,$$

$$\begin{vmatrix} \alpha \frac{\tau}{\kappa} - \mu - \lambda & 0 \\ 0 & -\kappa - \lambda \end{vmatrix} = 0, \quad (31)$$

$$\therefore \lambda_1 = \frac{\alpha\tau - \mu\kappa}{\kappa}, \lambda_2 = -\kappa.$$

We see that λ_2 is clearly negative but λ_1 can be positive or negative. If we consider the conditions mentioned in the subsection “Equilibrium points”, λ_1 must be positive, otherwise we cannot find a positive interior point. Hence, the dynamical system is unstable at the equilibrium point $E_1 = (0, 0)$.

Theorem 2. (Local Stability Theorem). The dynamical system is asymptotically stable at the point E_2 if $\alpha\tau + \mu\sigma > \mu(\xi + \omega)$ and $\delta + \omega + \xi > \sigma$ otherwise unstable.

Proof 5. For proof see Appendix A. $\square$

4.4. Global Stability

Theorem 3. (Global Stability Theorem). The dynamical system is globally unstable at the equilibrium point $E_2 = ((\alpha\tau - \mu\kappa/\gamma\kappa), \tau/\kappa)$ where $\kappa = \delta - \sigma + \omega + \xi$.

Proof 6. For proof see Appendix B. $\square$

5. Various Qualitative Appraisals of the Model

5.1. Determining the Existence of a Limit Cycle

Theorem 4. Model system (3) and (4) does not exhibit any limit cycle in the interior of the positive quadrant of the PB - plane.

Proof 7. Let, $E(P, B) = 1/PB$. Clearly, $E(P, B)$ is positive in the interior of the positive quadrant of the PB - plane.

Taking as

$$\begin{aligned} e_1(P, B) &= \alpha BP - \gamma P^2 - \mu P, \\ e_2(P, B) &= \tau + \sigma B - \xi B - \omega B - \delta B. \end{aligned} \quad (32)$$

Then

$$\Delta(P, B) = \frac{\partial}{\partial P}(e_1 E) + \frac{\partial}{\partial B}(e_2 E),$$

$$\therefore \Delta(P, B) = \frac{\gamma}{B} - \frac{\tau}{PB^2}. \quad (33)$$

Based on the above equation, we see that all the parameters and state variables $P(t)$ and $B(t)$ are positive for $t \geq 0$. Consequently, we can deduce that the sign of Δ remains constant and it is not equal to zero inside the positive quadrant of the PB - plane. Therefore, according to the concept cited in Daga et al. [28]; the model system can't possess any limit cycle inside the interior of the PB -plane quadrant. $\square$

5.2. Characteristics of States Equilibrium Values with Respect to α . We will discuss the characterization of the equilibrium values of the population of Japan and birth rate based on Misra and Singh [29]. From equations (3) and (4), we obtain two functions of P^* , B^* and α in below:

$$f(P^*, B^*, \alpha) = \alpha P^* B^* - \gamma P^{*2} - \mu P^*,$$

$$g(P^*, B^*, \alpha) = \tau + \sigma B^* - \xi B^* - \omega B^* - \delta B^*,$$

$$\therefore \frac{dP^*}{d\alpha} = \frac{\begin{vmatrix} (\partial f(P^*, B^*, \alpha)/\partial B^*) & (\partial f(P^*, B^*, \alpha)/\partial \alpha) \\ (\partial f(P^*, B^*, \alpha)/\partial P^*) & (\partial g(P^*, B^*, \alpha)/\partial B^*) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \alpha)/\partial P^*) & (\partial g(P^*, B^*, \alpha)/\partial P^*) \\ (\partial g(P^*, B^*, \alpha)/\partial P^*) & (\partial g(P^*, B^*, \alpha)/\partial B^*) \end{vmatrix}},$$

$$\Rightarrow \frac{dP^*}{d\alpha} = \frac{P^* B^*}{\mu + 2\gamma P^* - \alpha B^*}. \quad (34)$$

It is clear that the numerator is positive always. Under the condition $2\gamma P^* > \alpha B^*$, We observe that the denominator is positive. Consequently $(dP^*/d\alpha) > 0$. Again,

$$\therefore \frac{dB^*}{d\alpha} = \frac{\begin{vmatrix} (\partial f(P^*, B^*, \alpha)/\partial \alpha) & (\partial f(P^*, B^*, \alpha)/\partial P^*) \\ (\partial g(P^*, B^*, \alpha)/\partial \alpha) & (\partial g(P^*, B^*, \alpha)/\partial P^*) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \alpha)/\partial P^*) & (\partial g(P^*, B^*, \alpha)/\partial B^*) \\ (\partial g(P^*, B^*, \alpha)/\partial P^*) & (\partial g(P^*, B^*, \alpha)/\partial B^*) \end{vmatrix}},$$

$$\Rightarrow \frac{dB^*}{d\alpha} = 0. \quad (35)$$

However, Figure 5 provides a visual representation of this constant phenomenon.

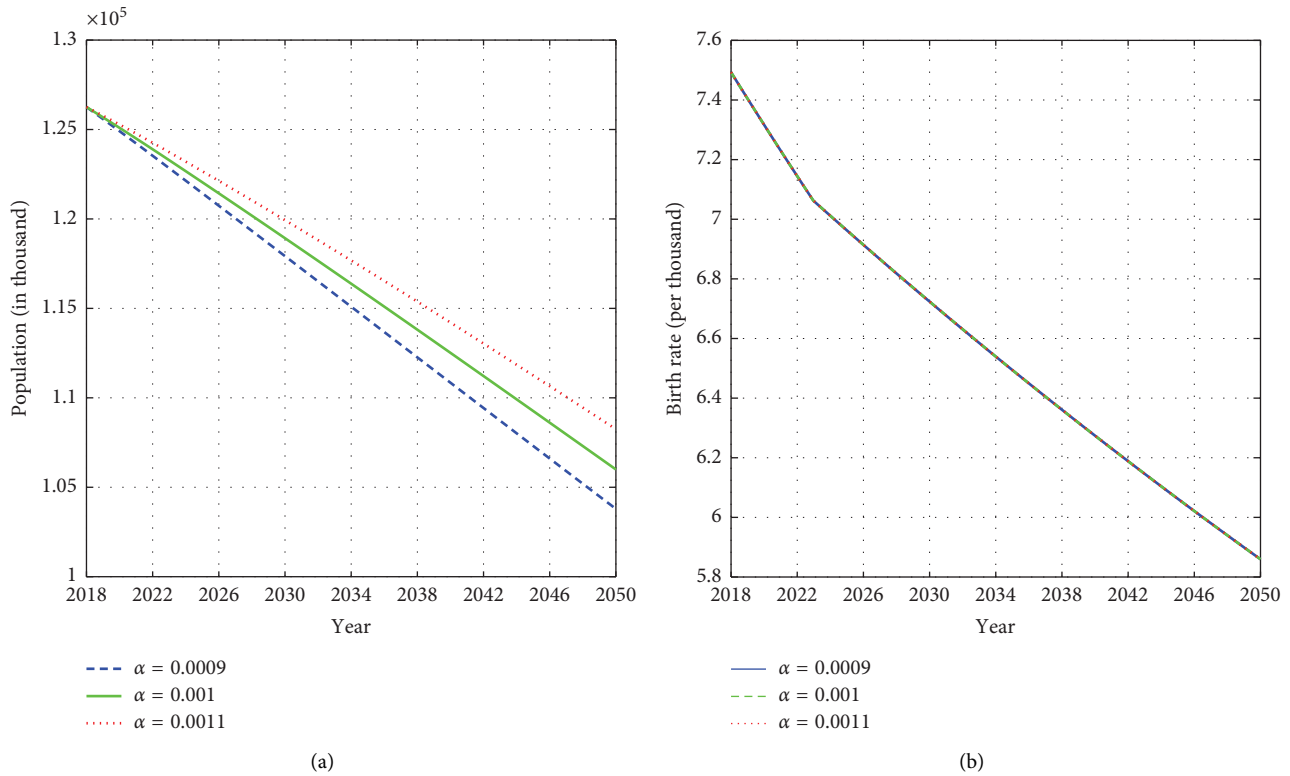


FIGURE 5: (a) More birth rate and immigration implies more population presence. (b) The birth rate is changing continuously from year to year constantly.

5.3. Characteristics of State's Equilibrium Values Relating to σ . This part will focus on the characteristics analysis of the equilibrium values of the Japanese population and birth rate according to Misra and Singh [29]. Two functions of P^* , B^* ,

and σ are obtained from equations (3) and (4) in the following:

$$f(P^*, B^*, \sigma) = \alpha P^* B^* - \gamma P^{*2} - \mu P^*,$$

$$g(P^*, B^*, \sigma) = \tau + \sigma B^* - \xi B^* - \omega B^* - \delta B^*,$$

$$\therefore \frac{dP^*}{d\sigma} = \frac{\begin{vmatrix} (\partial f(P^*, B^*, \sigma)/\partial B^*) & (\partial f(P^*, B^*, \sigma)/\partial \sigma) \\ (\partial g(P^*, B^*, \sigma)/\partial P^*) & (\partial g(P^*, B^*, \sigma)/\partial B^*) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \sigma)/\partial P^*) & (\partial f(P^*, B^*, \sigma)/\partial B^*) \\ (\partial g(P^*, B^*, \sigma)/\partial P^*) & (\partial g(P^*, B^*, \sigma)/\partial B^*) \end{vmatrix}}, \quad (36)$$

$$\Rightarrow \frac{dP^*}{d\sigma} = \frac{\alpha P^* B^*}{(\mu + 2\gamma P^* - \alpha B^*)(\xi + \delta + \omega - \sigma)}.$$

It is clear that the numerator is positive always. Under the previous conditions, we can observe that the denominator will be positive. Consequently $(dP^*/d\sigma) > 0$. Again,

$$\begin{aligned} \therefore \frac{dB^*}{d\sigma} &= \frac{\begin{vmatrix} (\partial f(P^*, B^*, \sigma)/\partial \sigma) & (\partial f(P^*, B^*, \sigma)/\partial P^*) \\ (\partial g(P^*, B^*, \sigma)/\partial \sigma) & (\partial g(P^*, B^*, \sigma)/\partial P^*) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \sigma)/\partial P^*) & (\partial f(P^*, B^*, \sigma)/\partial B^*) \\ (\partial g(P^*, B^*, \sigma)/\partial P^*) & (\partial g(P^*, B^*, \sigma)/\partial B^*) \end{vmatrix}}, \\ \Rightarrow \frac{dB^*}{d\sigma} &= \frac{B^*}{\xi + \delta + \omega - \sigma}. \end{aligned} \quad (37)$$

$$f(P^*, B^*, \delta) = \alpha P^* B^* - \gamma P^{*2} - \mu P^*,$$

$$g(P^*, B^*, \delta) = \tau + \sigma B^* - \xi B^* - \omega B^* - \delta B^*,$$

$$\begin{aligned} \therefore \frac{dP^*}{d\delta} &= \frac{\begin{vmatrix} (\partial f(P^*, B^*, \delta)/\partial B^*) & (\partial f(P^*, B^*, \delta)/\partial \delta) \\ (\partial g(P^*, B^*, \delta)/\partial B^*) & (\partial g(P^*, B^*, \delta)/\partial \delta) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \delta)/\partial P^*) & (\partial f(P^*, B^*, \delta)/\partial B^*) \\ (\partial g(P^*, B^*, \delta)/\partial P^*) & (\partial g(P^*, B^*, \delta)/\partial B^*) \end{vmatrix}}, \\ \Rightarrow \frac{dP^*}{d\delta} &= \frac{-\alpha P^* B^*}{(\mu + 2\gamma P^* - \alpha B^*)(\xi + \delta + \omega - \sigma)}. \end{aligned} \quad (38)$$

It is evident that the numerator is always negative. However, given the prior circumstances, it is evident that the denominator will be positive. Therefore, $(dP^*/d\delta) < 0$. Again,

$$\begin{aligned} \therefore \frac{dB^*}{d\delta} &= \frac{\begin{vmatrix} (\partial f(P^*, B^*, \delta)/\partial \delta) & (\partial f(P^*, B^*, \delta)/\partial P^*) \\ (\partial g(P^*, B^*, \delta)/\partial \delta) & (\partial g(P^*, B^*, \delta)/\partial P^*) \end{vmatrix}}{\begin{vmatrix} (\partial f(P^*, B^*, \delta)/\partial P^*) & (\partial f(P^*, B^*, \delta)/\partial B^*) \\ (\partial g(P^*, B^*, \delta)/\partial P^*) & (\partial g(P^*, B^*, \delta)/\partial B^*) \end{vmatrix}}, \\ \Rightarrow \frac{dB^*}{d\delta} &= \frac{-B^*}{\xi + \delta + \omega - \sigma}, \end{aligned} \quad (39)$$

which suggest us to conclude that $(dB^*/d\delta) < 0$. Biologically, we can say that, as a result of rising living expenses and the difficulty for couples to raise a family budget, Japan's growing rate of price inflation lowers the birth rate. Due to fewer births than there are aging people, this leads to a diminishing human population. However, this continual incidence has been revealed visually in Figure 7.

5.5. Behaviour of the Population if the Birth Rate Monotonically Climbs. The dynamics of the population will next be examined in light of the monotonically rising birthrate.

Now, it can be inferred from the preceding criteria that $(dB^*/d\sigma) > 0$. Nevertheless, this ongoing occurrence can be shown graphically in Figure 6.

5.4. Properties of Equilibrium Values of States in Regard to δ . In this part, we delve into the examination of the equilibrium values of the Japanese population and birth rate, drawing insights from Misra and Singh [29]. The analysis centers around two key functions, namely f and g for P^* (representing the equilibrium population), B^* (indicating the equilibrium birth rate), and δ (a parameter of significance). These functions are derived from equations (3) and (4).

This signifies that the number of birthrate must be higher than in the past.

Then from equation (3), we get

$$\frac{dP}{dt} = \alpha BP - \gamma P^2 - \mu P. \quad (40)$$

Now, we differentiate both sides with respect to time t

$$\begin{aligned} \frac{d^2 P}{dt^2} &= \alpha P \frac{dB}{dt} + \alpha B \frac{dP}{dt} - 2\gamma P \frac{dP}{dt} - \mu \frac{dP}{dt} \\ \Rightarrow \frac{d^2 P}{dt^2} &> \alpha B \frac{dP}{dt} - 2\gamma P \frac{dP}{dt} - \mu \frac{dP}{dt}. \end{aligned} \quad (41)$$

Because of the fact that birthrate is monotonically increasing, $\alpha P(dB/dt)$ must be a positive term.

Again for the critical points or equilibrium points,

$$\frac{dP}{dt} = \frac{dB}{dt} = 0. \quad (42)$$

Then, we find that, at the equilibrium point

$$\frac{d^2 P}{dt^2} > 0. \quad (43)$$

Therefore, the population will tend to be at its lowest at the critical point. This means that once the birth rate begins to rise, the population will begin to steadily expand.

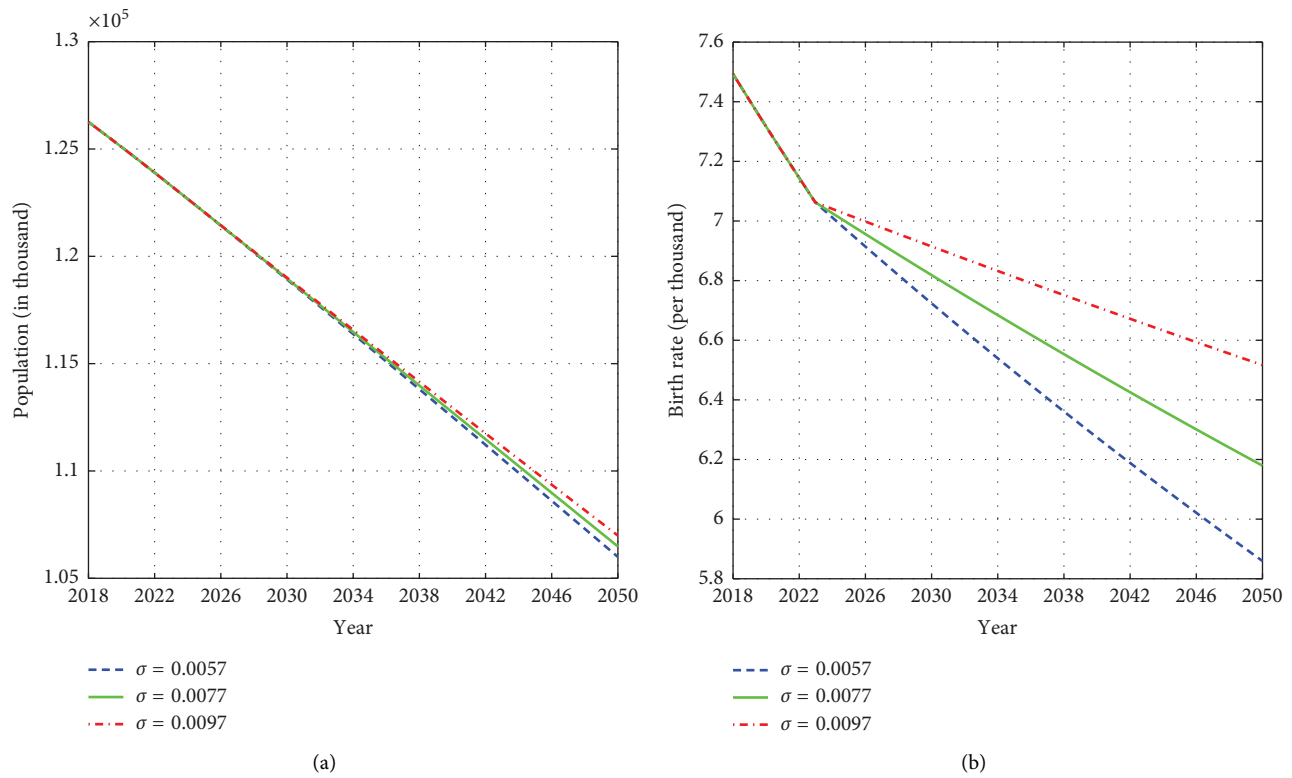


FIGURE 6: (a) Increased incentives help to slow down population decline. (b) It is possible to increase the birth rate by offering more financial incentives to have a baby.

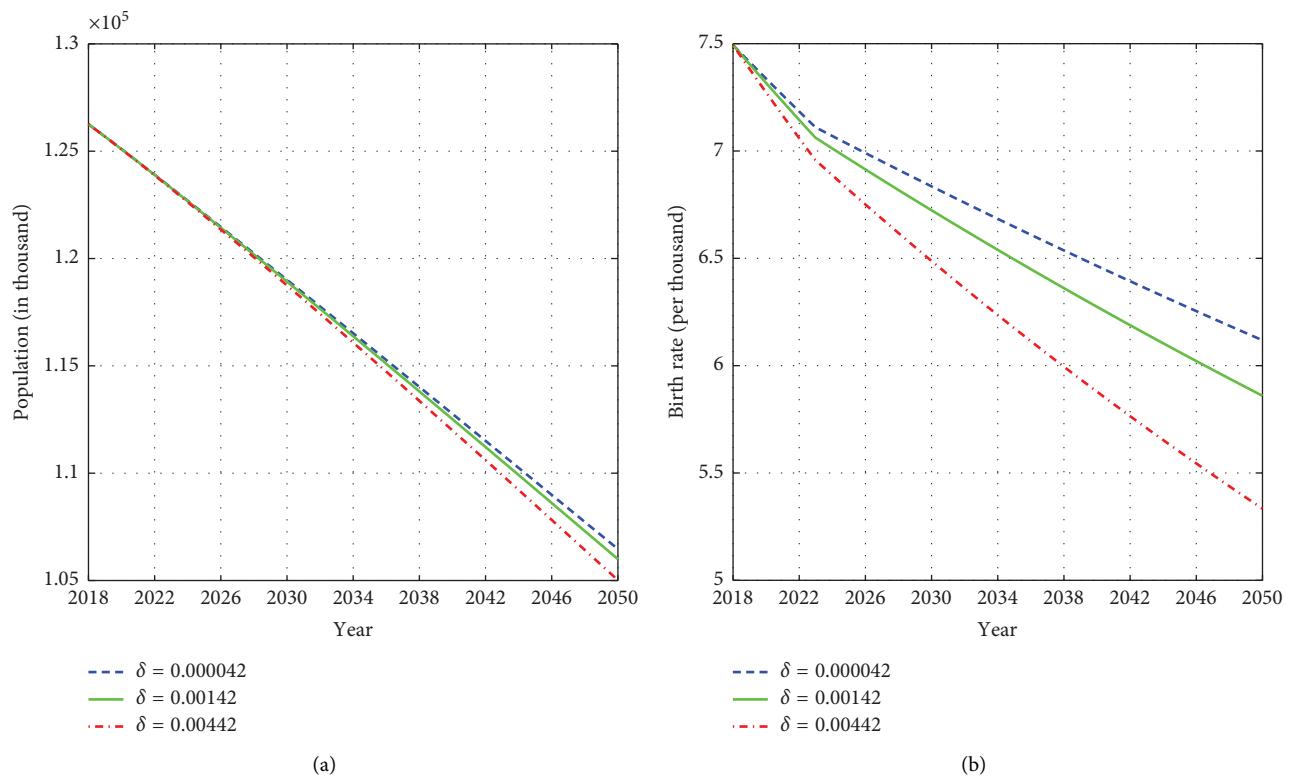


FIGURE 7: (a) Increased price inflation implies a greater decline in the population. (b) The more the price inflation is, the less the birth rate is.

6. Parameters Estimations

Here, we have utilized the procedure of the least square curve fitting method (see Figure 8) for the proposed model described in equations (3) and (4) to have the realistic values of the parameters involved and are shown in Table 1. In this regard, we have some of the parameters values that can be achieved or estimate; such include the natural death rate and birth rate. The reported cases here we consider are from 2018 to 2023 for the data fitting [3].

7. Sensitivity Analysis

The sensitivity index is a measure that reflects the degree to which a certain parameter is significant in this model that represents the increase in population in Japan. A procedure that is similar to the one described was used to compute the sensitivity index parameter for the future status of the Japanese population (R_0).

As a function of the differentiable parameter β , the normalised forward sensitivity index of the variable M is defined as

$$\mathcal{K}_\beta^M = \frac{\partial M}{\partial \beta} \times \frac{\beta}{M}. \quad (44)$$

The sensitivity indices are expected to vary from -1 to 1. Specifically,

$$\begin{aligned} \mathcal{K}_\alpha^{R_0} &= \frac{\partial R_0}{\partial \alpha} \times \frac{\alpha}{R_0} = 1, \\ \mathcal{K}_\tau^{R_0} &= \frac{\partial R_0}{\partial \tau} \times \frac{\tau}{R_0} = 1, \\ \mathcal{K}_\delta^{R_0} &= \frac{\partial R_0}{\partial \delta} \times \frac{\delta}{R_0} = -0.1121, \\ \mathcal{K}_\omega^{R_0} &= \frac{\partial R_0}{\partial \omega} \times \frac{\omega}{R_0} = -0.3275, \\ \mathcal{K}_\xi^{R_0} &= \frac{\partial R_0}{\partial \xi} \times \frac{\xi}{R_0} = -0.6551, \\ \mathcal{K}_\sigma^{R_0} &= \frac{\partial R_0}{\partial \sigma} \times \frac{\sigma}{R_0} = 0.0947, \\ \mathcal{K}_\gamma^{R_0} &= \frac{\partial R_0}{\partial \gamma} \times \frac{\gamma}{R_0} = -5 \times 10^{-8}, \\ \mathcal{K}_\mu^{R_0} &= \frac{\partial R_0}{\partial \mu} \times \frac{\mu}{R_0} = -1. \end{aligned} \quad (45)$$

A number of factors affect the sensitivity index in our system. The sensitivity index values for each parameter, as determined by the parameter value, are shown in Table 2. There is a visual representation of Table 1 in Figure 9.

8. Numerical Simulations

Numerical analysis has been performed on the proposed model using MATLAB (R2018b). The fourth-order Runge–Kutta algorithm has been used to graphically portray the interaction between state variables and model parameters and to display phase portraits for the appropriate parameters. In order to solve the model represented by equations (3) and (4), we have used the initial values of $P(0) = 126255.866$ (in thousands) and $B(0) = 7.493$ (per thousand), together with the parameter values obtained from statistical data provided in Table 1.

If the current trend seen from 2018 and 2023 persists, the solution of the model has been represented by Figure 10. Figure 10(a) illustrates that by 2050, the population will fall to 104,900 (in thousands), while the birth rate will be lowered to 5.146, as indicated in Figure 10(b).

On the other side, if the birth rate is increased to $\tau = 0.0081$ and $\sigma = 0.0057$ by incentive and awareness programs after 2023 respectively, the resulting situation is shown in Figure 11. In this scenario, by the year 2050, the birthrate is expected to be 5.877 (refer to Figure 11(b)). At that time, the population is projected to be 1.06×10^5 (in thousands), which aligns with the predictions made by The World Counts [30].

Besides, in the following subsection, the variation in population and birth rate dynamics (illustrated in Figure 11) with different factors has been pointed out.

8.1. The Dynamics of Parametric Variation for State Variables. The variations in the state variables with regard to δ are shown in Figure 7, where δ signifies the reduction in birth rate resulting from price inflation. Population decline is shown in Figure 7(a) for all δ values up to about the year 2027. However, the current population has seen significant effects from the value δ after this year. If $\delta = 0.00442$ is high, the population in 2050 will be nearly 1.05×10^5 thousand, but lower values of $\delta = 0.00142, 0.000042$ reveal 1.06×10^5 and 1.07×10^5 thousand, respectively, indicating that the population is decreasing at a faster rate with larger δ values.

Again, the rate of declining birth rate for all values of δ is quite close up to 2023, as shown in Figure 7(b). However, after 2023, the birth rate falls starkly with a slowdown in inflation. This is because individuals will have more challenges in providing for their families owing to the increased cost of items in the market. It has been noted that when the value of δ was just 0.000042, the birth rate in 2050 would be 6.119. On the other side, when the value of δ is increased to 0.00442, the birth rate would be decreased by over 87% compared to the previous value.

Besides, similar patterns in population and birth rate have been found in Figure 12, like to the preceding one. It implies that when the value of ξ increases, indicating a higher level of business activity, the birth rate decreases more. This decrease in birth rate has been observed as well to have a proportional impact on the population of Japan.

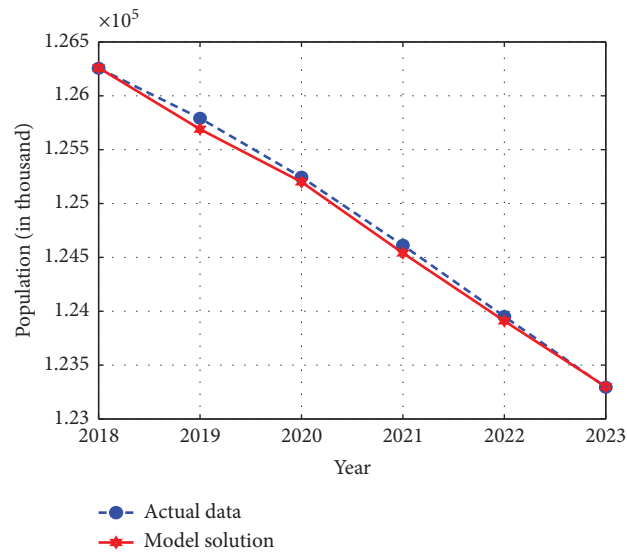


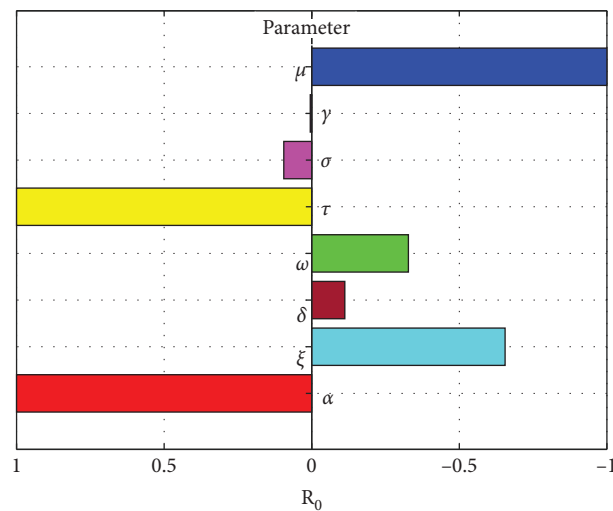
FIGURE 8: Data versus model fitting, 2018 till 2023.

TABLE 1: Parameter value and their descriptions.

Symbol	Parameter's description	Value (year ⁻¹)
α	The increasing rate of population based on by immigration and birth rate	0.001
γ	Rate of logistic growth for Japan human population	3.0×10^{-10}
μ	Natural death rate of human population	0.012
ξ	The declining birth rate caused by workaholism (business in work)	0.0083
ω	The falling birth rate due to technological progress	0.00415
δ	Decreasing birth rate caused by price inflation	0.00142
σ	The rising birth rate as a result of incentives	0.0012
τ	The rising birth rate by making awareness program	0.0056

TABLE 2: Sensitivity Indices of R_0 with parameters.

Parameter	Sensitivity index	Parameter	Sensitivity index
α	1	ξ	-0.6551
τ	1	σ	0.0947
δ	-0.1121	γ	-5×10^{-8}
ω	-0.3275	μ	-1

FIGURE 9: Effects of parameter variations on R_0 .

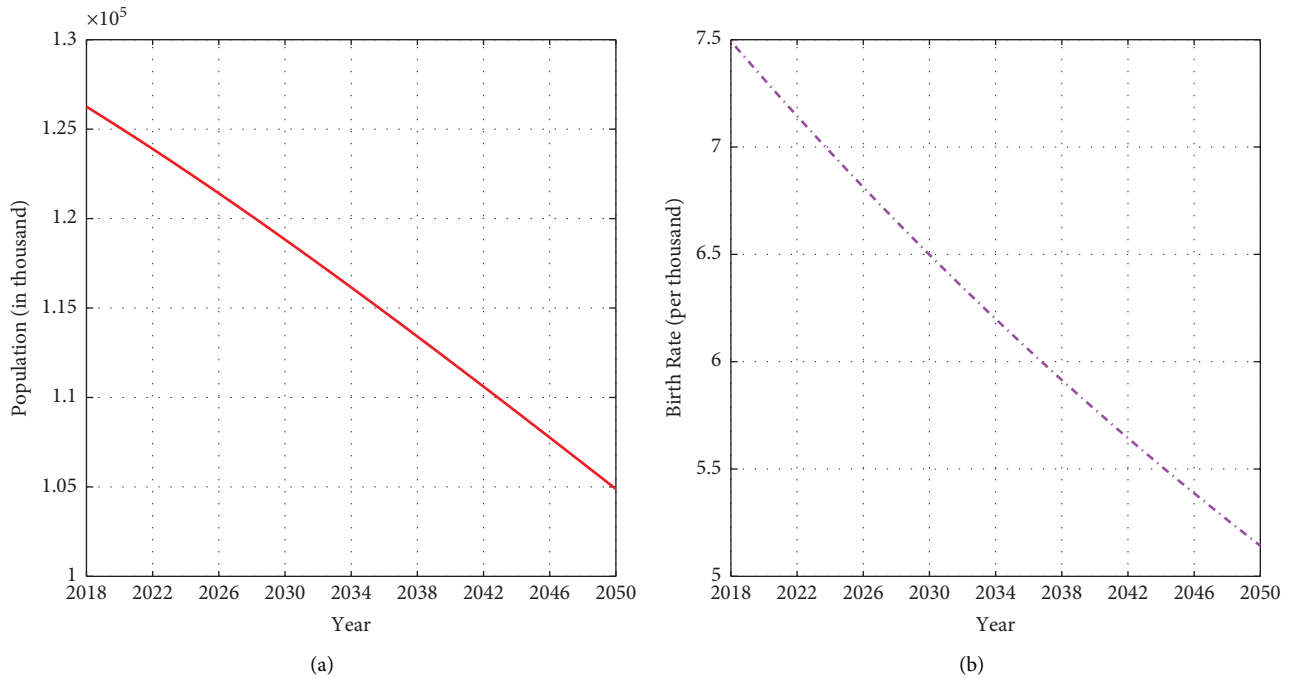


FIGURE 10: (a) The population is seeing a linear decline over time. (b) As more and more time passes, the birth rate continues to fall.

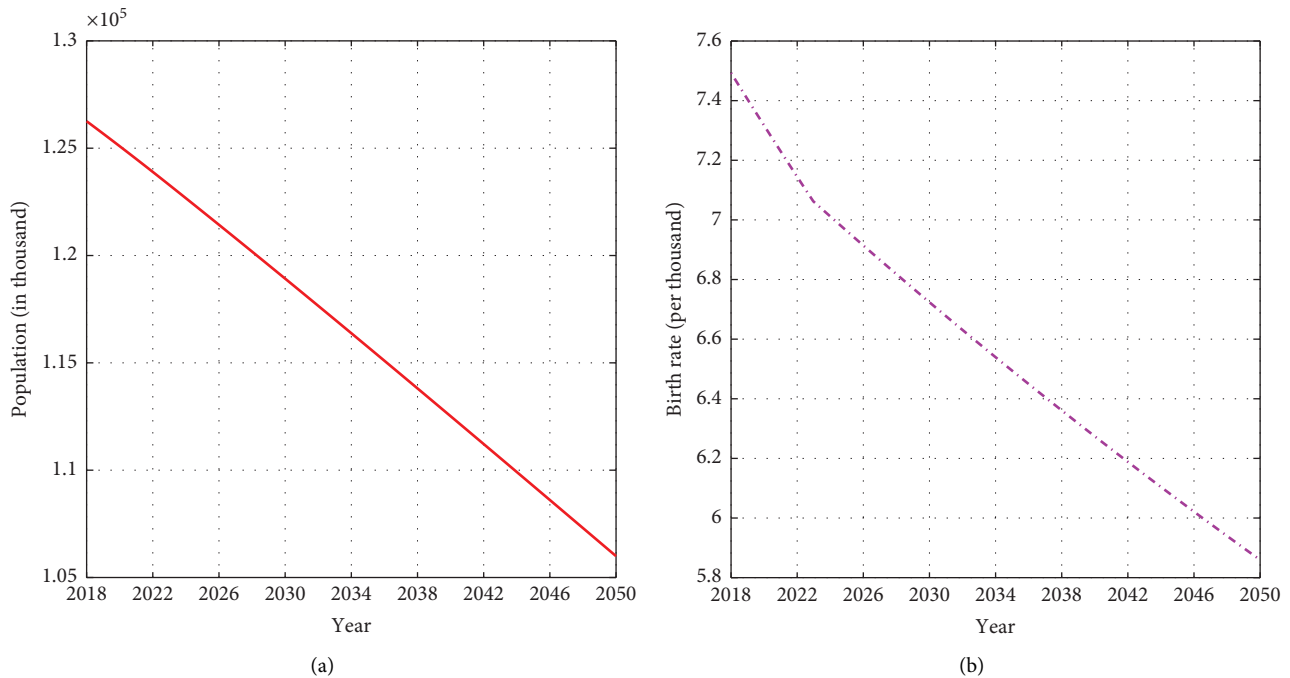


FIGURE 11: (a) The population has undergone a slow but consistent decrease over time. (b) As a result of increased incentives and awareness programs to have a family, the birth rate slowed around 2023 after first falling linearly.

On the flip side, Figure 6 illustrates the alterations in the state variables related to σ , demonstrating the increase in birth rate as a consequence of incentive provision. In Figure 6(a), it is evident that there exists a variation in the reduction of population that is independent of the value of σ . This variation is seen virtually up to the year 2030. However,

after that, the population steadily falls with a lesser style as the value of σ increases.

Also, it can be stated from Figure 6(b) that the birth rate remains constant up to 2023 when considering various values of σ . Once this period ends and awareness programs and incentive schemes are expanded beyond 2023,

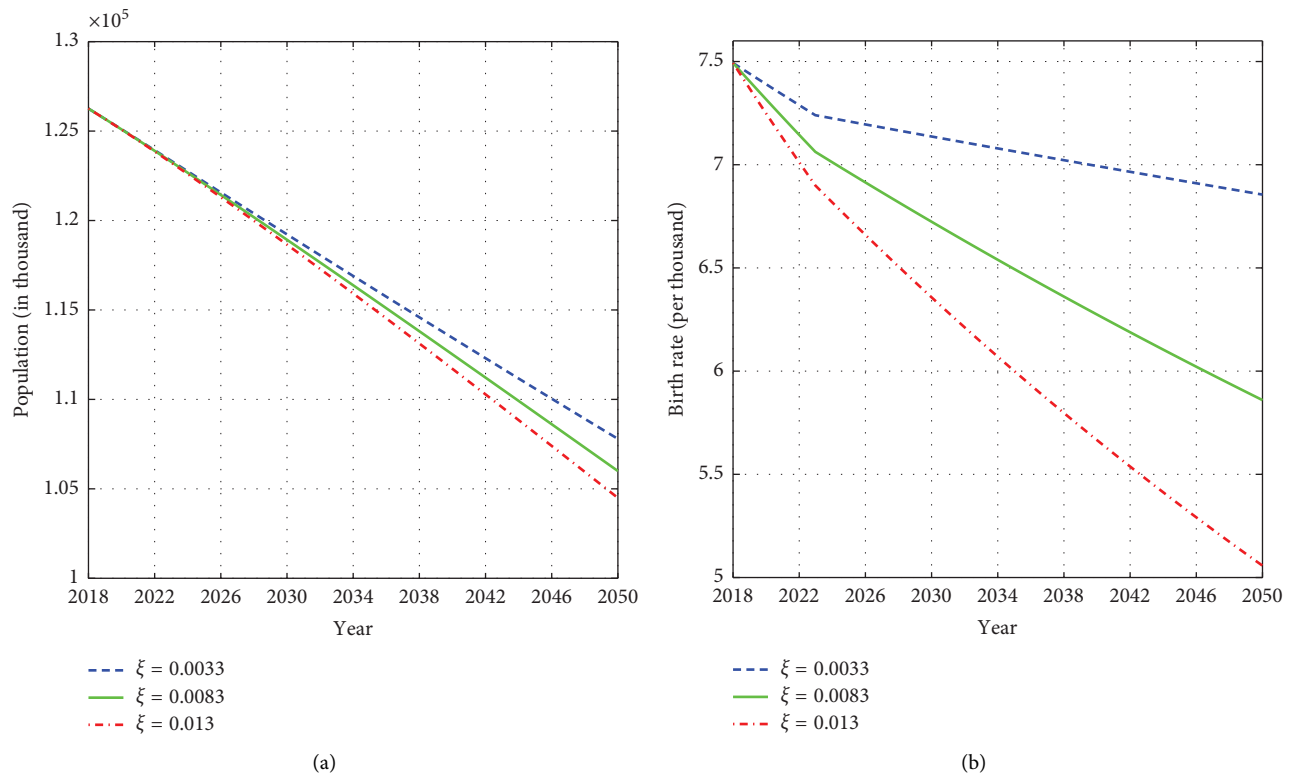


FIGURE 12: (a) The population is declining at a faster rate when there is more busyness in work. (b) When the duration of time spent working is increased, the birth rate falls.

a considerable variance in the birth rate with varying values of σ would be seen. During this period, the birth rate diminishes to 6.517 with increasing σ values of 0.0057. Contrary to the preceding one, the other two greater σ values show proportionally steeper declines.

Furthermore, Figure 5 provides a visual representation of how the state variables vary concerning the parameter α . In this particular context, the parameter α represents the rate at which the population improves as a result of the birth rate as well as immigration. Referring to Figure 5(a), Starting from the beginning of time, the population is changing in relation to α .

However, the gap between them has grown wider as time has progressed. The image shows that the population declines gradually with each α value, but that the decline accelerates dramatically from 2026 due to a decreased birth rate and immigration effect and continues until 2050. In addition, as can be shown in Figure 5(b), the birth rate consistently declines as α increases. Starting in 2018, it was 7.493 and by 2023, it had dropped to 7.094, a remarkable reduction in only four years. After that point, however, the decline slows down, and the birth rate would reach 5,861 by the end of 2050. It follows that the change in the birth rate does not rely on α .

8.2. Relationship Between State Variables. This phase portrait, which can be viewed in Figure 13, presents a description of the qualitative behavior of our model via

simulations. More specifically, it focuses on the maintenance of an alike or distinct level of incentives and the organization of awareness programs beyond the year 2023.

Figure 13(a) illustrates a steady decrease in population from the start to the final point, while maintaining a constant level. During this period, the birth rate decreases by 68% compared to the birth rate in the first year, over a span of 32 years. Moreover, in Figure 13(b), it is clarified that by implementing enhanced incentives and awareness programs starting in 2023, the rate of population decline has decreased compared to the level that was previously observed. This time, over a time frame of 32 years, the population reduction has reached 84%, with the birth rate lowering from 7.493 to 5.893.

The major factors that contribute to the increase or decrease of the Japanese population are highlighted in Figure 14, which shows the dynamic behavior of Japan's future population status. This status is a significant factor influencing the levels needed to maintain demographic balance, sustain economic growth, and ensure support for an ageing population. Figure 14(a) demonstrates a linear relationship between the rise in R_0 and the increase the rate of awareness program (τ) for promoting a more informed understanding of work-life balance, and available support systems. When looking at Figure 14(b), it is seen that the value of R_0 goes down slowly as the value of γ increases from 0 to 0.2. This is the opposite of what occurred in the previous scenario. Following that, the value of R_0 drops quickly proportional to the rate at which the value of γ grows.

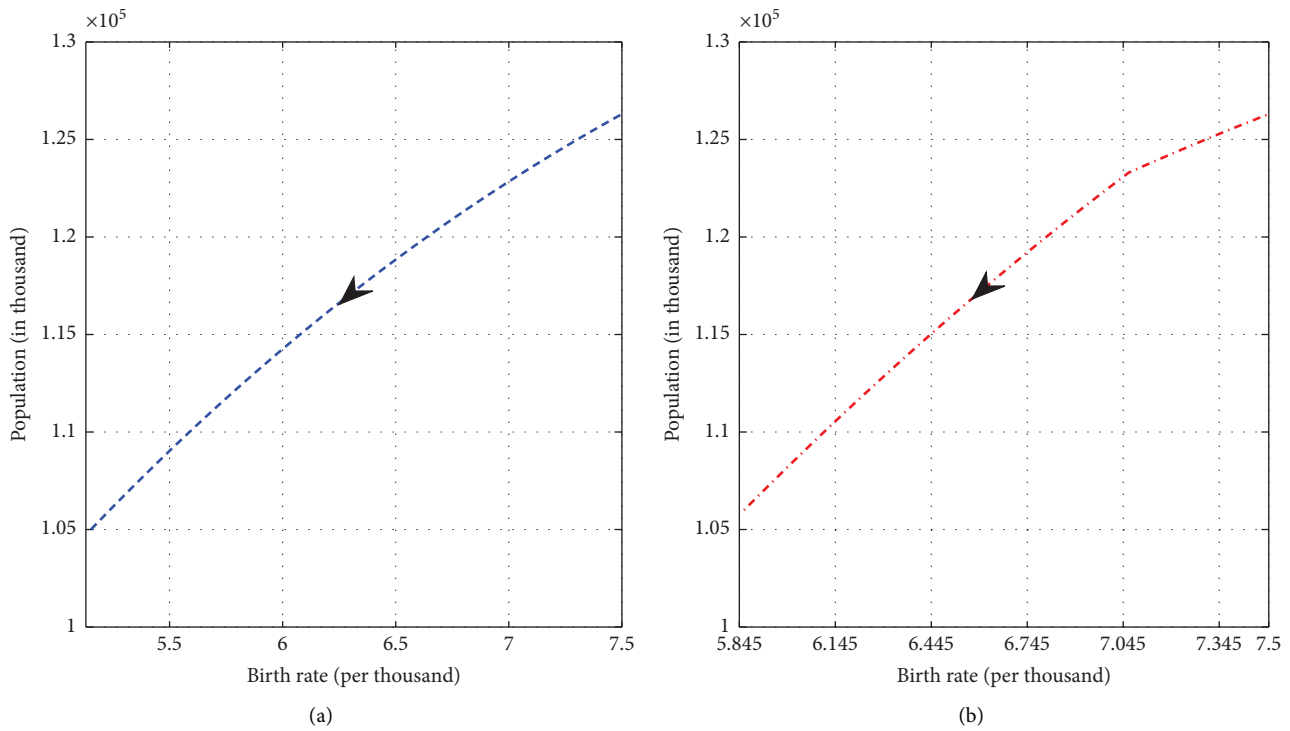


FIGURE 13: (a) Keeping financial aid and awareness programs always at their previous levels leads to a continuously falling population as a result of a decreased birth rate. (b) Introducing after 2023, the population experiences a drop slowly due to a lower birth rate brought about by enhanced incentives and awareness programs.

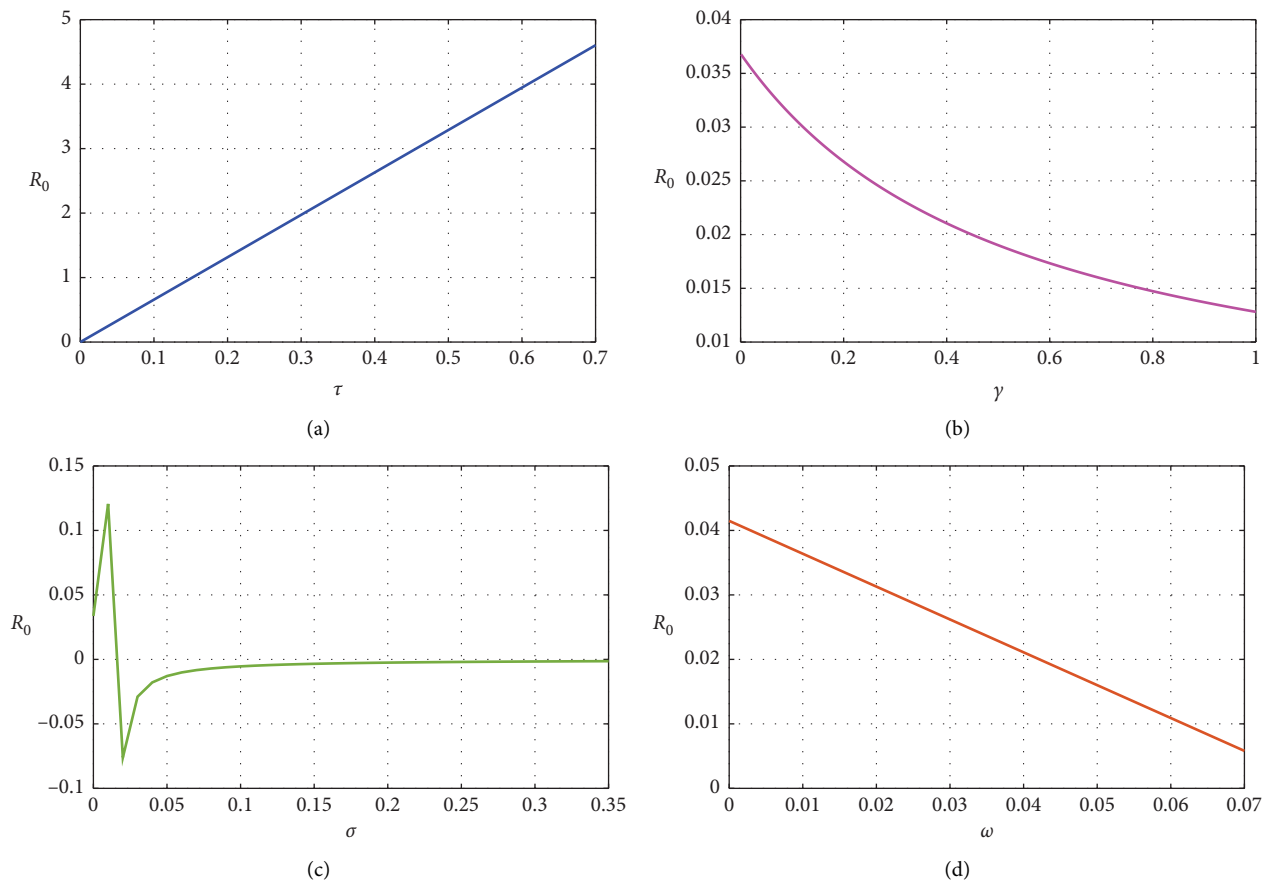


FIGURE 14: (a) As the program's level of awareness increases, the value of R_0 rises linearly. (b) The value of R_0 falls steadily as the value of γ gets higher. (c) While the value of σ elevates linearly R_0 , it eventually declines, yet broadening the value of σ preserves a constant R_0 . (d) The advancement of technology diminishes linearly the inclination towards forming families, as well as the value of the reproductive rate (R_0).

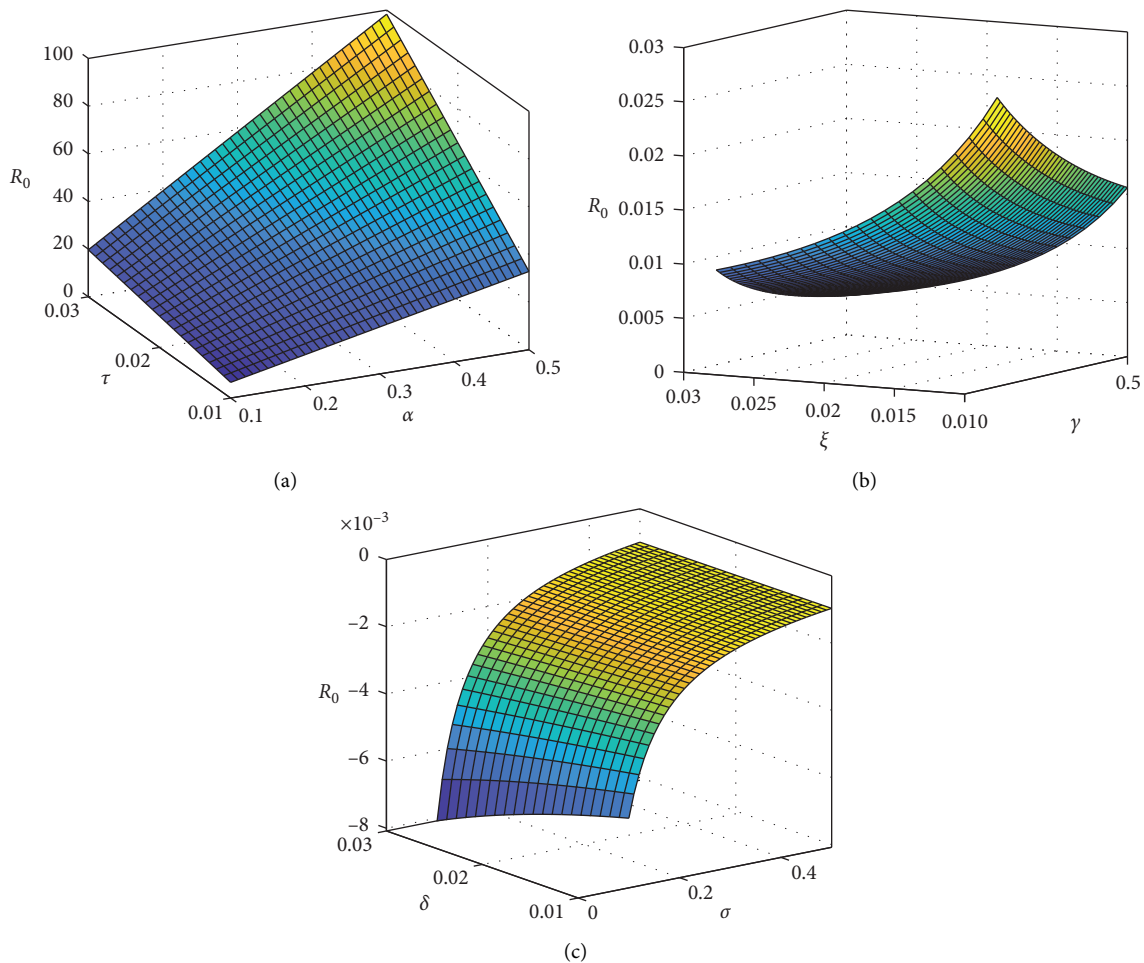


FIGURE 15: (a) The wide range of α and τ values has been used to highlight the influence of R_0 . (b) The consequence of R_0 has been graphically displayed for various values of γ and ξ . (c) The fluctuation of the values for R_0 can be observed with the value of incentives and price inflation changes.

In light of the fact that, a very interesting fluctuation has been detected in the value of R_0 concerning τ , as witnessed in Figure 14(c).

This graph (see Figure 14(c)) demonstrates that the value of R_0 first increases and remains positive as the value of σ increases. However, over time, this scenario undergoes a complete reversal. The value of R_0 has decreased and turned negative due to the potential difficulties posed by economic expansion and the provision of additional incentives. Nevertheless, with time, the effectiveness of these incentives may wane, leading to a reduction in population rise. Afterward, it continued climbing again as σ rose in value, and eventually, all values of σ contributed equally to the value of R_0 .

In addition, the value of R_0 points out a linear drop with ω (the lowering birth rate owing to technical advancement) as the technological progress grows which attractiveness causes the people to resort to family planning illustrated in Figure 14(d).

In the end, Figure 15 reflects the phase picture illustrating a relationship between various sensitive parameters and the future state of the population. If the value of R_0 is less

than 1, this situation would pose a risk. Conversely, if the value of R_0 is more than 1, it would contribute to a rise in population growth in the future. In doing this action, we can reduce the menacing impact of the declining birth rate in Japan.

9. Results and Discussion

Developed nations like Japan are facing new challenges related to aging and depopulation, compared to the developing world's tendency towards growing populations. There is an emerging concern about the potential for social imbalance and a decrease in the quality of life for the elderly caused by inadequate care and assistance when the senior population exceeds the younger people. Hence, Japan, being an island nation, is confronted with a substantial danger arising from its burgeoning older population.

In this study, the findings from Figures 10 and 11 indicate that if the rates of providing incentives and conducting awareness programs can be raised after 2023 compared to the period of 2018–2023, then the decline in birth rate will not be as rapid as it was before.

Increased price inflation in Japan lowers the birth rate by straining families financially and discouraging them from expanding owing to rising living expenses and economic uncertainty. As the expense of living increases, people may postpone or skip family planning, lowering the birthrate and worsening demographic issues. Figure 7 highlights this phenomenon.

At the same time, individuals in Japan can face a difficult situation in which they are unable to prioritize their family life due to the prevalent culture of lengthy working hours through heightened busyness in work and restricted work-life balance. This activity that is tied to employment works as a barrier to family planning, which leads to a fall in the birth rate and contributes to an overall decline in the population viewed in Figure 12.

On the reverse side, Figure 6 suggests that financial difficulties faced by prospective parents may be reduced by the implementation of financial incentives by the Japanese government and other organizations, which in turn makes family planning more accessible, potentially leading to a rise in the birth rate.

Finally, if we are able to succeed in increasing the birth rate by a variety of tactics, then the population drop will be achievable to a low level, as shown in Figure 5. This Figure also indicates that the arrival of young immigrants has the potential to address the declining population gap in Japan.

Furthermore, the future status of the population may increase linearly if more awareness programs an increased because by reducing social stigmas and misunderstandings about family planning and encouraging a more positive outlook on motherhood, a well-planned awareness campaign in Japan may help the country's population grow. Again, the position of the Japanese population might fall, nevertheless, if future intraspecies competition for food resources gets fiercer. The optimum level of providing incentives for adopting family planning, on the other hand, may contribute to future population increase. There is a constant fall in population status for the future in the case of improved technological advancement. All of this is shown in the specified sequence in Figures 14(a), 14(b), 14(c), and 14(d). Besides, Figure 15 illustrates the consequences that occur for the future population when factors such as varying levels of work business, price inflation, technological processes, and an awareness program, incentives are taken into consideration in three-dimensional environments.

Overall, it has been observed from Figures 10 and 11 that the population is also decreased after being given incentives offered by the government that describes the real situation as it is aligned to the recent document in Lim [18]. Besides, as more as the price of the product falls, as more as the birth rate can be raised visualized in Figure 7. At the same time, if price inflation decreases then people will reduce their engagement in busyness in work, and also Figure 12 shows that spending less time in work can rise in birth rate. Then, in this situation, incentives can be helpful for the government to increase the population of a country. Thus, it is suggested that combination of the strategies—controlling price

inflation to reduce price of the product in the market, less spend time on the workplace along with giving incentives—can be more effective than the only giving incentives offered by the government.

10. Conclusions

Japan, a country with the world's third-largest economy, is experiencing an unprecedented population catastrophe with significant social, economic, and political consequences due to a declining birthrate and a fast population. To study this burgeoning problem, the research has put out a mathematical model to depict the dynamics of birthrate in conjunction with population growth. Subsequently, both theoretical and numerical evaluations have been performed on this model. Using the positivity and boundedness theorems, we have concluded our theoretical analysis that our model has a unique solution and that it is biologically valid. The future status of the Japanese population provides insights into whether the population is projected to grow or decline. According to several real-world relevant criteria, the first equilibrium point E_1 is unstable, whereas the second one, E_2 , is globally unstable but locally asymptotically stable. Among other qualitative studies, the proposed model does not expose any limit cycle. Afterward, we conducted several tests to determine the features, and one of them is for σ (the rate of providing incentives), we found that both $(dP^*/d\sigma) > 0$ and $(dB^*/d\sigma) > 0$. Again, a sensitivity analysis was performed using the values obtained from parameter estimation, which allowed for the identification of the parameters that show their sensitive nature. After that, numerical simulations were performed out, which revealed that if the trend of the population dropping continues, then Japan would be faced with a significant challenge by the year 2050 due to the lower presence of young people as a result of a reduction in the birth rate. Regarding this matter, as the workload increases, there is a gradual rise in price inflation. Additionally, the growing reliance on technology leads to a more rapid decline in the birth rate. Furthermore, immigration can show a positive impact on the population rise in Japan. It is recommended to raise awareness about the decreasing number of babies born in Japan and provide substantial incentives from the government and other social development organizations. Additionally, efforts should be made to control price inflation. Failure to address this ongoing decline in birth rate and population could pose an alarming risk to the survival of the country. Therefore, more incentives and awareness campaigns including price deflation and reduced-load work may boost the birth rate of Japan. As part of future work, optimal control would be used to optimize the cost of incentives and arrange awareness programs.

Appendix A: Proof of Theorem 2

At the equilibrium point $E_2 = ((\alpha\tau - \mu\kappa/\gamma\kappa), \tau/\kappa)$ where $\kappa = \delta - \sigma + \omega + \xi$

The Jacobin matrix becomes

$$\Rightarrow J|_{E_2} = \begin{pmatrix} \alpha \frac{\tau}{\kappa} - 2\gamma \frac{\alpha\tau - \mu\kappa}{\gamma\kappa} - \mu & \alpha \frac{\alpha\tau - \mu\kappa}{\gamma\kappa} \\ 0 & -\kappa \frac{\tau}{\kappa} \end{pmatrix}. \quad (\text{A.1})$$

Therefore, the characteristic equation is

$$\begin{vmatrix} \alpha \frac{\tau}{\kappa} - 2\gamma \frac{\alpha\tau - \mu\kappa}{\gamma\kappa} - \mu - \lambda & \alpha \frac{\alpha\tau - \mu\kappa}{\gamma\kappa} \\ 0 & -\tau - \lambda \end{vmatrix} = 0, \quad (\text{A.2})$$

After solving the above equation, we get two values

$$\lambda_1 = -\kappa, \lambda_2 = -\frac{(\alpha\tau - \mu\kappa)}{\kappa}. \quad (\text{A.3})$$

Within this context along with previous conditions, all the eigenvalues are negative. As a result, the point of equilibrium, denoted by E_2 , is asymptotically stable.

Appendix B: Proof of Theorem 3

Proof B.1. In order to investigate the model's global stability, we constructed the following Lyapunov function in accordance with Ahmad et al. [31] for model (3) and (4).

$$\Omega(t) = \left[P - P^* - \frac{1}{P^*} \ln\left(\frac{P}{P^*}\right) \right] + \left[B - B^* - \frac{1}{B^*} \ln\left(\frac{B}{B^*}\right) \right], \quad (\text{B.1})$$

clearly at the equilibrium point E_2 , the function $\psi(t) = 0$ for the prescribed values of $P(t) = P^*$ and $B(t) = B^*$. Furthermore, the given function is strictly positive, whenever $P > P^*$ and $Q > Q^*$. $\square$

Now, the time derivative of the function $\Omega(t)$ results in

$$\begin{aligned} \frac{d\Omega(t)}{dt} &= \frac{dP}{dt} + \frac{dB}{dt} - \frac{(dP/dt)}{P} - \frac{(dB/dt)}{B} \\ &= (\alpha BP - \gamma P^2 - \mu P) + (\tau + \sigma B - \xi B - \omega B - \delta B) \\ &\quad - (\alpha B - \gamma P - \mu) - \left(\frac{\tau}{B} + \sigma - \xi - \omega - \delta \right) \\ &= \alpha B(P - 1) + \gamma P(1 - P) + \mu(1 - P) + \tau \left(1 - \frac{1}{B} \right) + \sigma_1(1 - B). \end{aligned} \quad (\text{B.2})$$

We see that $(d\Omega/dt)$ is positive.

Therefore, E_2 is globally unstable by satisfying Lyapunov–Lasalle's invariance principle [32].

Data Availability Statement

The data used to support the findings of this study are included in the article.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Pulak Kundu: conceptualization, formal analysis, methodology, data curation, software, writing—original draft, writing—review and editing.

Uzzwal Kumar Mallick: conceptualization, methodology, software, writing—review and editing, supervision, validation.

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